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3.2

$\delta(q_1, 0) = (q_2, x, R)$
$\delta(q_1, 1) = (q_3, x, R)$
$\delta(q_1, \#) = (q_8, R)$
$\delta(q_2, 0) = (q_2, R)$
$\delta(q_2, 1) = (q_2, R)$
$\delta(q_2, \#) = (q_4, R)$
$\delta(q_4, x) = (q_4, R)$
$\delta(q_4, 0) = (q_6, x, L)$
$\delta(q_6, 0) = (q_6, L)$
$\delta(q_6, 1) = (q_6, L)$
$\delta(q_6, x) = (q_6, L)$
$\delta(q_6, \#) = (q_7, L)$
$\delta(q_7, 0) = (q_7, L)$

$\delta(q_7, 1) = (q_7, L)$
$\delta(q_7, x) = (q_1, R)$
$\delta(q_8, x) = (q_8, R)$
$\delta(q_8, \sqcup) = (q_{accept}, R)$
$\delta(q_3, 0) = (q_3, R)$
$\delta(q_3, 1) = (q_3, R)$
$\delta(q_3, \#) = (q_5, R)$
$\delta(q_5, x) = (q_5, R)$
$\delta(q_5, 1) = (q_6, x, L)$
$\delta(q_6, 0) = (q_6, L)$
$\delta(q_6, 1) = (q_6, L)$
$\delta(q_6, x) = (q_6, L)$

a. 11 , b. 1#1 , c. 1##1 , d 10#11 , e 10#10

a. $(q_1, \triangleright \underline{1}1) \vdash_M (q_3, \triangleright x \underline{1})$
$\vdash_M (q_3, \triangleright x 1 \underline{\sqcup})$
$\vdash_M (q_{reject}, \triangleright x 1 \underline{\sqcup})$

b. $(q_1, \triangleright \underline{1}\#1) \vdash_M (q_3, \triangleright x \underline{\#} 1)$
$\vdash_M (q_5, \triangleright x \# \underline{1})$
$\vdash_M (q_6, \triangleright x \underline{\#} x)$
$\vdash_M (q_7, \triangleright \underline{x} \# x)$
$\vdash_M (q_1, \triangleright x \underline{\#} x)$
$\vdash_M (q_8, \triangleright x \# \underline{x})$
$\vdash_M (q_8, \triangleright x \# x \underline{\sqcup})$
$\vdash_M (q_{accept}, \triangleright x \# x \sqcup \underline{\sqcup})$

(c) $(q_1, \triangleright \underline{1}\#\#1) \vdash_M (q_3, \triangleright x \underline{\#}\#1)$
$\vdash_M (q_5, \triangleright x \# \underline{\#} 1)$
$\vdash_M (q_{reject}, \triangleright x \# \underline{\#} 1)$

(d) $(q_1, \triangleright \underline{1}0\#11) \vdash_M (q_3, \triangleright x \underline{0} \# 11)$
$\vdash_M (q_3, \triangleright x 0 \underline{\#} 11)$
$\vdash_M (q_5, \triangleright x 0 \# \underline{1} 1)$
$\vdash_M (q_6, \triangleright x 0 \underline{\#} x 1)$
$\vdash_M (q_7, \triangleright x \underline{0} \# x 1)$
$\vdash_M (q_7, \triangleright \underline{x} 0 \# x 1)$
$\vdash_M (q_1, \triangleright x \underline{0} \# x 1)$
$\vdash_M (q_2, \triangleright x x \underline{\#} x 1)$
$\vdash_M (q_4, \triangleright x x \# \underline{x} 1)$
$\vdash_M (q_4, \triangleright x x \# x \underline{1})$
$\vdash_M (q_{reject}, \triangleright x x \# x \underline{1})$

(e) $(q_1, \triangleright \underline{1}0\#10) \vdash_M (q_3, \triangleright X\underline{0}\#10)$

$\vdash_M (q_3, \triangleright X0\underline{\#}10)$

$\vdash_M (q_5, \triangleright X0\#\underline{1}0)$

$\vdash_M (q_6, \triangleright X0\underline{\#}X0)$

$\vdash_M (q_7, \triangleright X\underline{0}\#X0)$

$\vdash_M (q_7, \triangleright \underline{X}0\#X0)$

$\vdash_M (q_1, \triangleright X\underline{0}\#X0)$

$\vdash_M (q_2, \triangleright XX\underline{\#}X0)$

$\vdash_M (q_4, \triangleright XX\#\underline{X}0)$

$\vdash_M (q_4, \triangleright XX\#X\underline{0})$

$\vdash_M (q_6, \triangleright XX\#\underline{X}X)$

$\vdash_M (q_6, \triangleright XX\underline{\#}XX)$

$\vdash_M (q_7, \triangleright X\underline{X}\#XX)$

$\vdash_M (q_1, \triangleright XX\underline{\#}XX)$

$\vdash_M (q_8, \triangleright XX\#\underline{X}X)$

$\vdash_M (q_8, \triangleright XX\#X\underline{X})$

$\vdash_M (q_8, \triangleright XX\#XX\underline{\sqcup})$

$\vdash_M (q_{accept}, \triangleright XX\#XX\sqcup\underline{\sqcup})$

3.8 Input alphabet $\Sigma = \{0, 1\}$

Tape alphabet $\Gamma = \{0, 1, X, Y, \sqcup\}$. X marked 1. Y marked 0.

Start state $q_1$; halting $q_{accept}$, $q_{reject}$

(a) $L_1 = \{w \mid w$ has an equal number of 0s and 1s$\}$

(1) Return to the leftmost of the tape.

(2) Scan right to find the first unmarked symbol:

① If it is only $(X, Y, \sqcup)$, ACCEPT.

② If it is an unmarked 0, overwrite with Y and go to step (3).

③ If it is an unmarked 1, overwrite with X and go to step (4).

(3) (Matched a 0) Return to leftmost, then scan right for the first unmarked 1.

① If found, mark it X and go to step 1.

② If you hit ⊔ without find 1, return REJECT.

(4) (Matched a 1) Return to leftmost, then scan right for the first marked 0.

① If found, marked it Y and go to step 1.

② If none found, REJECT.

(b) $\{w \mid w$ contains twice as many 0s as 1s$\}$

(1) Return to the leftmost.

(2) Scan right for the unmarked 1:

① If none exists, go to step (5).

② If found, mark it X and continue.

(3) Return to the leftmost. Scan right for the unmarked 0:

① If found, mark it Y.

② If not found, REJECT.

(4) Return to the leftmost. Scan right for the unmarked 0:

① If found, mark it Y.

② If not found, REJECT.

(5) When no 1 remains, scan the tape:

① If any 0 remains, REJECT.

② Otherwise, ACCEPT.



(c) $\{w \mid w$ does not contain twice as many 0s as 1s$\}$

(1) We can use (b) turing machine as base model $M_b$.

① If $M_b$ ACCEPT, the (c) turing machine REJECT.

② If $M_b$ REJECT, the (c) turing machine ACCEPT.

3.15 Assume $A, B$ are decidable by deciders $D_A, D_B$.
Input $w$.

a. union, $A \cup B$

(1) Run $D_A(w)$. If it accepts, return ACCEPT.
(2) Otherwise, run $D_B(w)$. If it accepts, ACCEPT; else REJECT.
So $D_A(w) \cup D_B(w)$ is still closed.

b. concatenation, $A \cdot B = \{xy : x \in A, y \in B\}$
For each $i = 1 \cdots n$, $x \in [1, i-1]$, $y \in [i, n]$
(1) Run $D_A(x)$, $D_B(y)$. If both accept, return ACCEPT.
(2) If all pairs are tested, it can not halt. Return REJECT.

d. complementation, $\bar{A}$
(1) Run $D_A(w)$.
(2) If $D_A(w)$ accepts, return REJECT; else ACCEPT.

e. Intersection $A \cap B$.
(1) Run $D_A(w)$. If it reject, return REJECT; else go to (2).
(2) Run $D_B(w)$. If it reject, return REJECT; else return ACCEPT.

3.16 Assume languages $L_1, L_2$ are Turing-recognizable with recognizers $M_1, M_2$.
Input is $w$.

a. Union, $L_1 \cup L_2$.

(1) If either $M_1(w)$ or $M_2(w)$ accepts, return ACCEPT.

(2) If both $M_1(w)$ and $M_2(w)$ rejects, return REJECT.

(3) If $M_1(w)$ loops and $M_2(w)$ rejects, this recognizer runs forever.

b. Concatenation. $L_1 \cdot L_2 = \{x, y : x \in L_1, y \in L_2\}$

(1) If $M_1(x)$ and $M_2(y)$ both are accept, return ACCEPT.

(2) If $M_1(x)$ and $M_2(y)$ both eventually halts and at least one reject, return REJECT. Otherwise, the machine may run forever.

d. Intersection $L_1 \cap L_2$

(1) If $M_1(x)$ and $M_2(y)$ both are accept, return ACCEPT.

(2) If $M_1(x)$ and $M_2(y)$ both eventually halts and at least one reject, return REJECT. Otherwise, the machine may run forever.